Milestone M2 Report

Detection-Certified Power Allocation for Cognitive ISAC Relaying
Under Compound-Gaussian Clutter and Imperfect CSI

Project category: Research Project
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Abstract

This report documents the state of the project at Milestone M2. Cognitive integrated sensing and communication (ISAC) relaying lets a dual-function base station (DBS) detect primary targets while relaying secondary users on the same spectrum. The state of the art protects the radar with a minimum signal-to-interference-plus-noise ratio (SINR), which guarantees detection only under Gaussian disturbance and perfect channel knowledge. We formulate the problem with the guarantee that a radar actually needs: a per-target detection probability $P_{\mathrm{D}} \geq \beta$ at a fixed false-alarm probability P_{FA}, under compound-Gaussian clutter and imperfect CSI. We have (i) a complete system model and problem formulation, (ii) a proof that turns the detection constraint into an exact bisection inside a sum-rate optimizer, (iii) a validated simulator that reproduces the baselines of the closest published work, and (iv) a 100-channel paired evaluation in that work's own setting. In that evaluation the proposed design kept every target at $P_{\mathrm{D}} \geq 0.9$ on 100% of channels with 7.78 bps/Hz, whereas the published SINR design reached a similar rate but protected only 10% of channels, and a worst-case robust design gave up 1.99 bps/Hz.

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1 M2 Summary and Checklist

Table 1 maps each M2 expectation to where it is addressed and to the artifact that evidences it.

Table 1: M2 expectations and where they are met.

M2 expectation	Section	Evidence
Synthesis of the literature (not paper summaries)	2	Thematic synthesis along four axes; comparison matrix (Table 2)
Clear technical gap statement	3	Boxed gap statement with the two failure mechanisms
Well-defined problem formulation	5	Signal, clutter, CSI and rate models; optimization problem (5)
Proposed system/model	4	System diagram (Fig. 1) and detection law (4)
Proposed methodology	6	Proposition 1, Algorithm 1, complexity, evaluation protocol
Technical artifact / working core	7	Python simulator (8 modules), validation suite, 4,200-point campaign, compiled 6-page IEEE paper
Baseline reproduction	7.3	Hu *et al.* non-robust, worst-case robust and relay baselines in their published setting
Initial evaluation	7.4	Paired 100-channel results, sweeps over β, ϵ, P_c, $P^{\max}$, N_r, N_t, clutter shape
SOTA positioning	8	What existing approaches achieve vs. what this project adds

2 Synthesis of the Relevant Literature

We organize the literature along four axes that together define the design space of a sensing-protected cognitive ISAC relay: *(A)* the network architecture, *(B)* the sensing metric used inside the optimization, *(C)* the clutter model, and *(D)* the treatment of channel uncertainty. The synthesis below shows that each axis is well studied on its own, but that the combinations needed for a guaranteed-detection cognitive relay are missing.

2.1 Axis A: Spectrum-sharing and relay-assisted ISAC architectures

Spectrum sharing between radar and communication was first approached through interference alignment and interference exploitation between separate systems [1, 2], and has since moved towards dual-function platforms, with surveys covering interference management, waveforms, reconfigurable surfaces and security [3, 4, 5, 6]. The cognitive-radio view, in which secondary access is admissible only while the primary function is preserved, was recently brought into relay-assisted ISAC by Hu *et al.* [7]: a multi-antenna DBS senses K targets and relays M secondary pairs with an amplify-and-forward (AF) matrix, and the secondary sum rate is maximized subject to a per-target sensing SINR floor, under perfect or norm-bounded CSI. Cognitive radar designs based on learning [8] balance detection against throughput but do not relay secondary traffic. *Takeaway:* the cognitive relay architecture exists, but its primary-protection rule is an SINR floor.

2.2 Axis B: How sensing enters the optimization

Four families of sensing metrics appear. (i) *SINR/SCNR* constraints or objectives are the most common because they are tractable, both in cognitive relaying [7] and in cell-free and distributed ISAC [9, 10, 11]. (ii) *Estimation metrics* such as the Cramér–Rao bound target parameter accuracy

rather than detection [12, 13]. (iii) *Information/divergence metrics* (e.g., KLD) act as detection surrogates under statistical uncertainty [14]. (iv) *Detection probability* itself appears in trade-off analyses [15, 16], in joint power-and-threshold design [17], and, most recently, as an explicit floor in ISAC resource allocation, motivated by the observation that CRB-only allocation can hide poor detection [18]. *Takeaway:* the field is moving from surrogate metrics towards explicit detection guarantees, but the existing detection-constrained designs assume Gaussian disturbance and, in [18], perfect CSI.

2.3 Axis C: Clutter modelling

Classical radar theory models heavy-tailed clutter as compound-Gaussian, a Gaussian speckle modulated by a random texture, and derives Neyman–Pearson and CFAR detectors for it [19]; adaptive detection under correlated inverse-Gamma texture is an active topic [20]. In ISAC, detection under compound-Gaussian clutter has been analysed together with input-distribution design [21], and a recent survey identifies non-Gaussian, heterogeneous clutter as a central open issue for clutter-aware ISAC [22]. Optimization papers, however, usually work with a clutter covariance or subspace and an SCNR metric [9, 10, 11]. *Takeaway:* heavy-tailed clutter is understood for detection, but it has not been carried into the constraint of a cognitive relay design, where an SINR floor is blind to the clutter distribution at fixed mean power.

2.4 Axis D: Channel uncertainty

Imperfect CSI is handled with norm-bounded worst-case models [7], with statistical or Bayesian error models [9, 14], and with noise-uncertainty-robust detectors [23]; detection–rate trade-offs with imperfect channel information are studied in [16]. Worst-case designs are safe but conservative, and nominal designs are efficient but fragile. A specific effect matters in the relay: a receive beamformer built from *estimated* channels nulls the estimate, not the estimation error, so the residual leakage is not visible to a nominal SINR constraint. *Takeaway:* no existing design calibrates the detection threshold and the power allocation to the CSI-error law in a relay.

2.5 Synthesis

Table 2 places representative works on the four axes. The pattern is consistent: the works that guarantee detection assume Gaussian disturbance and/or perfect CSI and do not relay secondary users; the works that model clutter or CSI uncertainty optimize a surrogate (SINR, SCNR, KLD); and the cognitive relay architecture protects the radar with an SINR floor.

Table 2: Comparison matrix of representative work (✓: addressed; ×: not addressed; –: not the focus).

Work	Architecture	Sensing guarantee	Clutter model	CSI	Cognitive relay
Hu *et al.* [7]	CR-ISAC AF relay	SINR floor	Gaussian	bounded	✓
Zhang *et al.* [21]	monostatic ISAC	P_{D} analysis, input design	compound-Gaussian	–	×
Mazahir, Tabassum [18]	multi-user ISAC BS	explicit P_{D} floor	Gaussian	perfect	×
Behdad *et al.* [9]	cell-free ISAC	sensing SINR	clutter (Gaussian)	imperfect	×
Rivetti *et al.* [10]	cell-free mmWave	SCNR	clutter subspace	imperfect	×
Zafari *et al.* [11]	distributed cell-free	SCNR	clutter	imperfect	×
Yoo *et al.* [14]	distributed ISAC	KLD	–	statistical	×
Obando *et al.* [17]	MIMO ISAC	detection error, threshold	Gaussian	–	×
Guo *et al.* [16]	MIMO-OFDM ISAC	P_{D}–rate trade-off	clutter-free	imperfect	×
This project	**CR-ISAC AF relay**	**$P_{\mathrm{D}} \geq \beta$ at fixed P_{FA}**	**compound-Gaussian**	**Gaussian error**	✓

3 Technical Gap Statement

Gap. In cognitive ISAC relaying, the primary radar is protected by a sensing-SINR floor [7]. An SINR floor fixes only the mean interference power. Under compound-Gaussian clutter and imperfect CSI it does not guarantee the detection probability that the radar is specified by, and no existing design replaces it with a false-alarm-calibrated detection guarantee that accounts for both effects.

Two mechanisms make the gap concrete:

1. *Clutter tail.* At equal mean power, a heavy-tailed texture raises the threshold needed for a given P_{FA}, so the SINR that once gave $P_{\mathrm{D}} = 0.9$ gives less.
2. *CSI-error leakage.* A receive beamformer built from estimated ST channels nulls the estimate. The nominal SINR row then becomes slack, the secondary users transmit at full power, and the unnulled estimation error degrades detection.

Research question. *How much secondary sum rate can a cognitive ISAC relay support when every target must meet $P_{\mathrm{D}} \geq \beta$ at a fixed P_{FA} under compound-Gaussian clutter and imperfect CSI, and how does this compare with SINR-floor designs in the published setting?*

4 Proposed System Model

We adopt the network of [7] (Fig. 1). A DBS with N_t transmit and N_r receive antennas serves K point targets at angles θ_k and M single-antenna secondary transmitter–receiver (ST–SR) pairs without a direct link. In slot 1 (uplink) the STs transmit to the DBS, which simultaneously emits a radar beam $\mathbf{v}$ ($\|\mathbf{v}\|^2 = P_s$) and receives the echoes. In slot 2 (downlink) it forwards its slot-1 observation through an AF matrix $\mathbf{W}$.

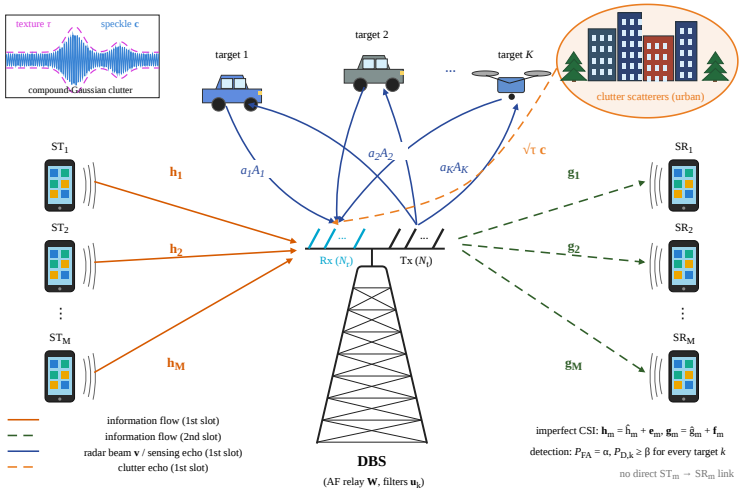


Figure 1: Proposed system: CR-enabled ISAC relaying under compound-Gaussian clutter. Secondary pairs communicate only through the DBS, which senses K targets in slot 1.

Slot-1 signal. The DBS receives

$$\mathbf{y} = \sum_{k=1}^{K} a_k \mathbf{A}_k \mathbf{v}\, s + \sum_{m=1}^{M} \sqrt{p_m}\, \mathbf{h}_m x_m + \mathbf{n} + \sqrt{\tau}\, \mathbf{c}, \tag{1}$$

with target response $\mathbf{A}_k = \beta_k \mathbf{b}(\theta_k)\mathbf{a}^T(\theta_k)$, ST power $0 \le p_m \le P_c$, noise $\mathbf{n} \sim \mathcal{CN}(\mathbf{0}, \sigma_n^2 \mathbf{I})$, and compound-Gaussian clutter: speckle $\mathbf{c} \sim \mathcal{CN}(\mathbf{0}, \sigma_c^2 \mathbf{I})$ with texture $\tau \sim \text{Gamma}(k, 1/k)$, constant over a coherent processing interval. The texture has unit mean, so $\sigma_r^2 = \sigma_n^2 + \sigma_c^2$ equals the noise power of [7]; a smaller shape k means heavier tails.

Imperfect CSI. The DBS knows only estimates and the error law: $\mathbf{h}_m = \widehat{\mathbf{h}}_m + \mathbf{e}_m$, $\mathbf{g}_m = \widehat{\mathbf{g}}_m + \mathbf{f}_m$, $a_k = 1 + e_k$, with $\mathbf{e}_m \sim \mathcal{CN}(\mathbf{0}, \epsilon \|\widehat{\mathbf{h}}_m\|^2 \mathbf{I}/N_r)$, $\mathbf{f}_m \sim \mathcal{CN}(\mathbf{0}, \epsilon \|\widehat{\mathbf{g}}_m\|^2 \mathbf{I}/N_t)$, $e_k \sim \mathcal{CN}(0, \epsilon)$. For worst-case baselines this Gaussian error is mapped to the norm-bounded model of [7] with the radius that contains it with probability 0.95.

Communication. SR_m receives $\mathbf{g}_m^T \mathbf{W}\mathbf{y} + n_m$. Treating the CSI-error terms as uncorrelated noise gives the achievable sum rate

$$R = \frac{1}{2} \sum_{m=1}^{M} \log_2\Big(1 + \frac{p_m |\widehat{\mathbf{g}}_m^T \mathbf{W} \widehat{\mathbf{h}}_m|^2}{\mathbb{E}|\mathbf{g}_m^T \mathbf{W}\mathbf{y}|^2 + \sigma^2 - p_m |\widehat{\mathbf{g}}_m^T \mathbf{W} \widehat{\mathbf{h}}_m|^2}\Big), \tag{2}$$

with average relay power $P_t = \mathbb{E}\|\mathbf{W}\mathbf{y}\|^2 \le P^{\max}$; both are available in closed form and reduce exactly to (4), (6), (7) of [7] at $\epsilon = 0$.

Detection law. For target k the DBS thresholds $T_k = \text{Re}\{e^{-j\angle \mu_k} \mathbf{u}_k^H \mathbf{y}\, s^*\}$ with unit-norm filter $\mathbf{u}_k$ and $\mu_k = \mathbf{u}_k^H \mathbf{A}_k \mathbf{v}$. Conditioned on the CSI errors and the texture, T_k is Gaussian with mean 0 ($\mathcal{H}_0$) or $|\mu_k|$ ($\mathcal{H}_1$), and with

$$V_k = I_k + \sum_{m=1}^{M} p_m |\mathbf{u}_k^H \mathbf{h}_m|^2 + \sigma_n^2 + \tau \sigma_c^2. \tag{3}$$

Averaging over the texture and the errors gives

$$P_{\text{FA}} = \mathbb{E}\, Q\Big(\frac{t_k}{\sqrt{V_k/2}}\Big) = \alpha, \qquad P_{\text{D}} = \mathbb{E}\, Q\Big(\frac{t_k - |\mu_k|}{\sqrt{(V_k + \epsilon|\mu_k|^2)/2}}\Big), \tag{4}$$

where the first equation fixes the threshold t_k under the true null law. In the Gaussian, perfect-CSI special case this reduces to $P_{\text{D}} = Q(Q^{-1}(\alpha) - \sqrt{2\Gamma_k})$, i.e., $P_{\text{D}} \ge \beta$ is the SINR floor $\eta(\beta) = \frac{1}{2}[Q^{-1}(\alpha) - Q^{-1}(\beta)]^2$ (9.80 dB for $\beta = 0.9$, $\alpha = 10^{-3}$), which explains why the published 10 dB floor is a good choice only in that special case.

5 Problem Formulation

We keep the objective and budgets of [7, (8)] and replace the SINR floor by the calibrated detection requirement:

$$\max_{\mathbf{p},\ \mathbf{W},\ \{\mathbf{u}_k\}} R(\mathbf{p},\mathbf{W}) \quad \text{s.t.} \quad 0 \le p_m \le P_c, \quad P_t(\mathbf{p},\mathbf{W}) \le P^{\max}, \quad P_{\mathrm{D},k}\big(t_k(\mathbf{p}),\mathbf{p}\big) \ge \beta \ \ \forall k. \tag{5}$$

- **Decision variables:** ST powers $\mathbf{p}$, relay matrix $\mathbf{W}$, receive filters $\{\mathbf{u}_k\}$; the radar beam $\mathbf{v}$ is fixed for all schemes.
- **Objective:** secondary sum achievable rate (2), averaged over the CSI error.
- **Constraints:** per-ST and relay power budgets; per-target $P_{\mathrm{D}} \ge \beta$ at $P_{\mathrm{FA}} = \alpha$ under (4).
- **Difficulty:** (5) is non-convex in $(\mathbf{p},\mathbf{W})$, and the detection constraint is an implicit, mixture-integral function of $\mathbf{p}$ with a threshold that itself depends on $\mathbf{p}$.

6 Proposed Methodology

6.1 Key result: the detection constraint is a monotone certificate

Proposition 1. *Fix $\mathbf{u}_k$, a direction $\mathbf{d} \ge \mathbf{0}$ and $\alpha < 1/2$. The calibrated threshold $t_k(s)$ for $\mathbf{p} = s\mathbf{d}$ increases strictly with s, and P_{D} decreases strictly with s whenever $t_k(s) < |\mu_k|$.*

Proof. Each p_m raises V_k in (3) almost surely, and $Q(t/\sqrt{V/2})$ increases with V, so t_k must grow to hold $P_{\mathrm{FA}} = \alpha$. The P_{D} integrand is $\Phi\big((|\mu_k| - t_k)/\sqrt{(V_k + \epsilon|\mu_k|^2)/2}\big)$; with $t_k < |\mu_k|$ its numerator falls and its denominator grows, and expectation preserves the strict decrease. □

Consequently the feasible scalings along $\mathbf{d}$ form an interval $[0, s_k^\star]$, found exactly by bisection, and the detection requirement enters the power update as one linear row per target, $\sum_m p_m \bar{c}_{km} \le s_k^\star \sum_m d_m \bar{c}_{km}$ with $\bar{c}_{km} = \mathbb{E}|\mathbf{u}_k^H \mathbf{h}_m|^2$, in the place of the SINR row of [7].

6.2 Algorithm

Algorithm 1 Detection-certified cognitive ISAC relay design

1: $\mathbf{p} \leftarrow$ largest common scaling of $P_c\mathbf{1}$ that meets $P_{\mathrm{D}} \ge \beta$; $\mathbf{W} \leftarrow$ MRC/MRT scaled to $P^{\max}$
2: **repeat**
3: $\quad \mathbf{u}_k \leftarrow$ MVDR filter loaded with the expected CSI-error covariance
4: $\quad$ for each target: bisection for $s_k^\star$ along $\mathbf{d} = \mathbf{p}$; form the certificate row
5: $\quad \mathbf{p} \leftarrow \arg\max R$ subject to the rows and $0 \le p_m \le P_c$ (relay gain rescaled to $P^{\max}$)
6: $\quad \mathbf{W} \leftarrow$ one WMMSE update of the relay (same KKT point as [7, Alg. 1])
7: **until** relative change of R below 10^{-4} (at most 40 iterations)
8: recompute $\mathbf{u}_k$; if a target fails, scale $\mathbf{p}$ down by bisection

Numerical integration. The expectations in (4) use generalized Gauss–Laguerre nodes for the texture and scrambled Sobol points for the CSI error. A design margin of 0.003 on β covers the largest quadrature error (0.0010) measured over 1,119 target cases.

Convergence and complexity. The power and relay steps never decrease R for fixed certificate rows; the rows are refreshed each iteration, so convergence is established empirically (Section 7.4), and a formal proof is planned work. Per iteration the cost is $\mathcal{O}(KN_r^3)$ for the filters, $\mathcal{O}(KBJGn_\tau)$

for the certificate (B bisection steps, J threshold steps, Gn_τ quadrature nodes) and $\mathcal{O}(N_t^3 N_r^3)$ for the relay step, the same order as the relay subproblem of [7]. Measured: 5.4 s per channel versus 2.7 s for the worst-case baseline (Python, one core).

6.3 Evaluation protocol

- **Setting:** the parameters of [7, Sec. VI]: $N_t = N_r = 8$, $M = K = 4$, 100 m links, path-loss exponent 4, $P_c = P^{\max} = 20$ dBm, $\sigma_r^2 = -80$ dBm. Added: $P_s = 20$ dBm, $|\beta_k|^2 = -90$ dB, $\sigma_c^2 = \sigma_n^2$, $k = 0.7$, $\epsilon = 0.01$, $\beta = 0.9$, $\alpha = 10^{-3}$.
- **Pairing:** 100 channel realizations per sweep point, identical for all schemes, so all differences are paired; 95% t-intervals and Wilcoxon signed-rank tests.
- **Fair evaluator:** every design is scored under the true clutter and CSI law, with its threshold calibrated to the true null; so all schemes meet $P_{\mathrm{FA}} = \alpha$ and differ only in P_{D} and rate.
- **Upper bound:** the proposed design evaluated with perfect CSI ($\epsilon = 0$), which shows how much rate the CSI error costs.
- **Metrics:** average sum rate; the share of channels on which every target meets $P_{\mathrm{D}} \geq \beta$ (*protected*); and the *protected sum rate*, which counts a channel's rate only when it is protected, mirroring the underlay rule.

7 Evidence of Progress

7.1 Technical artifact: working implementation

Table 3: Implemented and executed components (Python 3.14, NumPy, SciPy).

Component	Status and role
`mimo_core.py`	System model (1), clutter, CSI errors, detection law (4), rate (2), WMMSE and power steps
`schemes.py`	Proposed design (Alg. 1), all baselines, common true-law evaluator
`validate.py`, `quadrature_margin.py`	Independent checks against brute-force simulation; margin sizing
`run_sweeps.py`	Paired Monte-Carlo campaign: 7 sweeps, 4,200 run-points, 26,500 scheme evaluations
`convergence.py`, `plot_hu_style.py`, `make_tex.py`	Figures in the format of [7]; all report numbers generated from data
Result Set 1 (scalar relay)	Uncertainty-aware likelihood-ratio detector with held-out verification
Paper draft	6-page IEEE conference paper, compiled

7.2 Validation of the core component

Table 4: Independent checks of the model reductions.

Check	Result
Detection law (4) vs. brute-force simulation of the full N_r-antenna signal (4×10^5 CPIs per hypothesis, 16 targets)	max $\lvert\Delta P_{\mathrm{D}}\rvert = 0.0010$, max $\lvert\Delta P_{\mathrm{FA}}\rvert = 0.00015$
Closed-form rate and relay power vs. 2×10^4 CSI-error draws	within 0.23% and 0.013%
Worst-case relay power (trust-region) vs. 2×10^5 random errors	never below random search
WMMSE relay step	rate never decreased
Design quadrature vs. dense reference	largest P_{D} overestimate 0.0010 $<$ margin 0.003

7.3 Baseline reproduction

We implemented the constraint sets of Hu *et al.* [7] in their published setting and solve them with the same optimizer as the proposed design: Algorithm 1 on nominal channels with the published 10 dB floor and with the Gaussian-equivalent floor $\eta(\beta)$; the worst-case robust design (Hu's worst-case sensing and relay-power constraints); and the ZF, MRC/MRT and LMMSE relay structures. As in [7], a perfect-CSI curve serves as the upper bound. The reproduced baselines show the qualitative behaviour reported in [7]: the sum rate increases with the ST budget and saturates (Fig. 4), increases with the relay budget and the number of receive antennas (Table 6), robust designs lie below non-robust ones in rate, and the optimized relay outperforms the fixed relay structures. These are Hu-style constraint sets under a shared solver, not the authors' code.

7.4 Initial evaluation

Table 5: Default point ($\beta = 0.9$, $\epsilon = 0.01$, $k = 0.7$, 20 dBm), 100 paired channels. Last column: proposed minus scheme in protected sum rate, 95% interval and Wilcoxon p.

Scheme	SAR (bps/Hz)	Prot. (%)	mean $\min_k P_{D,k}$	5th pct.	Prot. SAR	Prop.−scheme, prot. SAR (Wilcoxon p)
Proposed (aware P_D certificate)	7.78 ± 0.16	100	0.917	0.903	7.78	—
Error-ignoring P_D design	7.68 ± 0.20	10	0.517	0.122	0.80	$+6.98 \pm 0.48$ ($< 10^{-4}$)
Hu Alg. 1, $\eta(\beta)$, non-robust	7.79 ± 0.17	9	0.432	0.072	0.72	$+7.05 \pm 0.46$ ($< 10^{-4}$)
Hu Alg. 1, $\eta = 10$ dB, non-robust	7.80 ± 0.17	10	0.441	0.082	0.80	$+6.98 \pm 0.48$ ($< 10^{-4}$)
Hu robust worst-case, $\eta(\beta)$	5.79 ± 0.15	100	1.000	0.999	5.79	$+1.99 \pm 0.12$ ($< 10^{-4}$)
Perfect CSI ($\epsilon = 0$), proposed	10.35 ± 0.29	100	0.974	0.903	10.35	-2.57 ± 0.25 ($< 10^{-4}$)

Table 5 is the headline result. The published 10 dB design reaches a raw rate (7.80 bps/Hz) similar to the proposed design (7.78 bps/Hz) but keeps all targets detectable on only 10% of channels; its rate is earned by failing the primary requirement. The worst-case design protects every target but gives up 1.99 ± 0.12 bps/Hz. The proposed design is the only one that protects every target and keeps the rate high: $+6.98 \pm 0.48$ bps/Hz of protected sum rate over the published design and $+1.99$ bps/Hz over the worst-case design ($p < 10^{-4}$ in both cases). Across all sweep points the proposed rule (including its perfect-CSI and structured-relay variants) protected 11200/11200 channel evaluations. With perfect CSI the proposed design reaches 10.35 bps/Hz; the remaining gap to 7.78 bps/Hz is the price of the CSI error, which the proposed design pays openly instead of hiding it in failed detections.

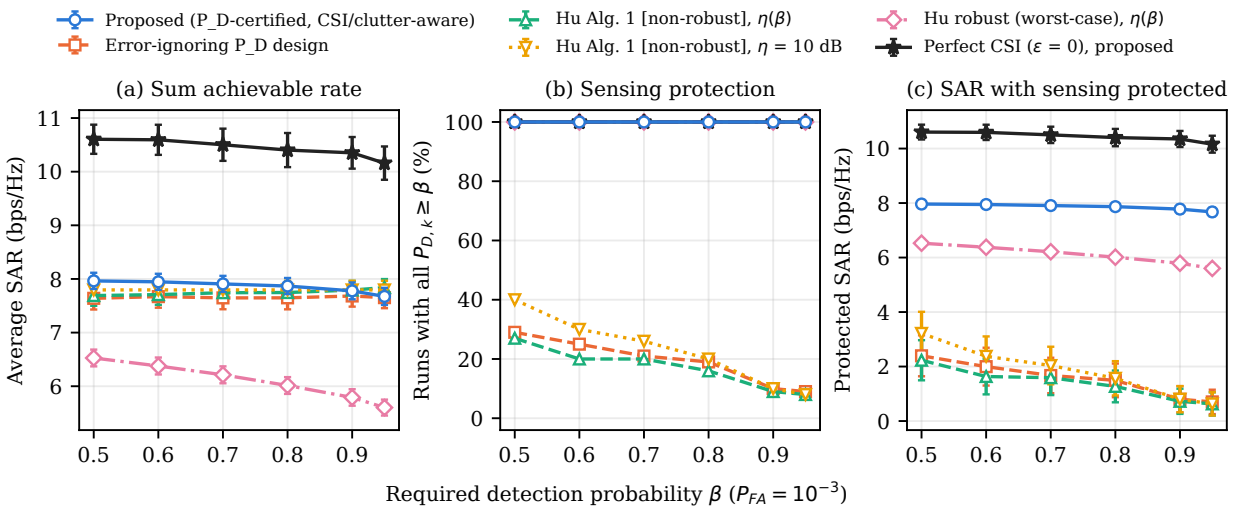


Figure 2: Versus the required P_{D}: (a) average sum rate, (b) channels with every target protected, (c) protected sum rate (counterpart of [7, Fig. 9]). The black curve is the perfect-CSI upper bound.

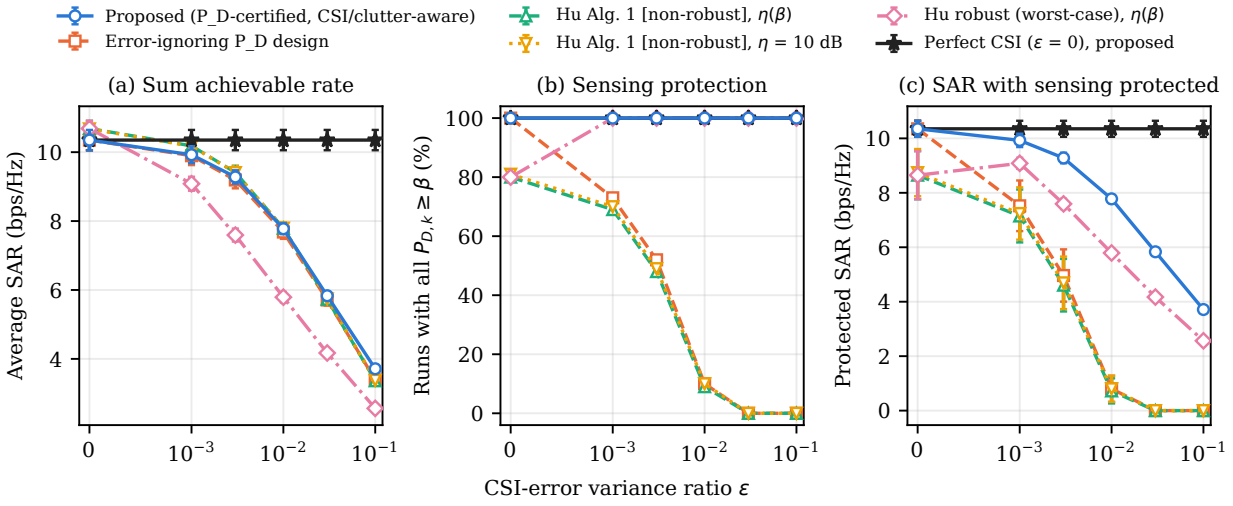


Figure 3: Versus the CSI-error ratio ϵ; Hu's radii 0.03–0.05 correspond to $\epsilon \approx 10^{-3}$.

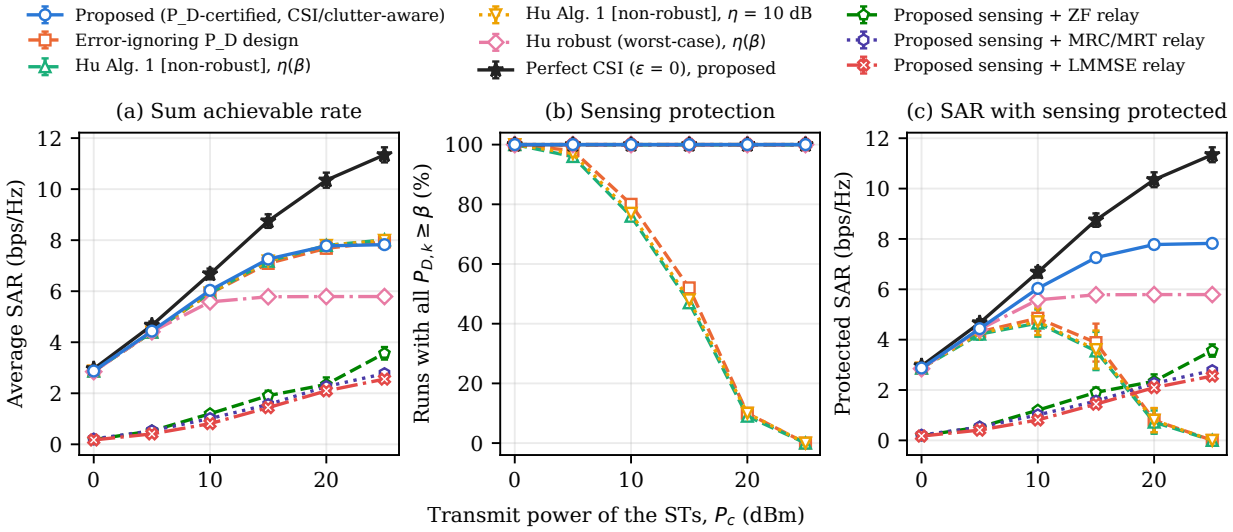


Figure 4: Versus the ST budget P_c, including the ZF, MRC/MRT and LMMSE relays (counterpart of [7, Fig. 3]).

Table 6: Protected sum rate in bps/Hz (% protected channels) at the ends of the budget and antenna sweeps.

Setting	Proposed	Hu, 10 dB	Hu robust
$P_c = 0$ dBm	2.87 (100)	2.87 (100)	2.85 (100)
$P_c = 25$ dBm	7.82 (100)	0.00 (0)	5.79 (100)
$P^{\max} = 15$ dBm	7.04 (100)	0.72 (10)	5.17 (100)
$P^{\max} = 30$ dBm	8.23 (100)	0.93 (11)	6.12 (100)
$N_r = 6$	4.95 (100)	1.89 (36)	3.65 (100)
$N_r = 16$	10.85 (100)	10.82 (100)	9.74 (100)

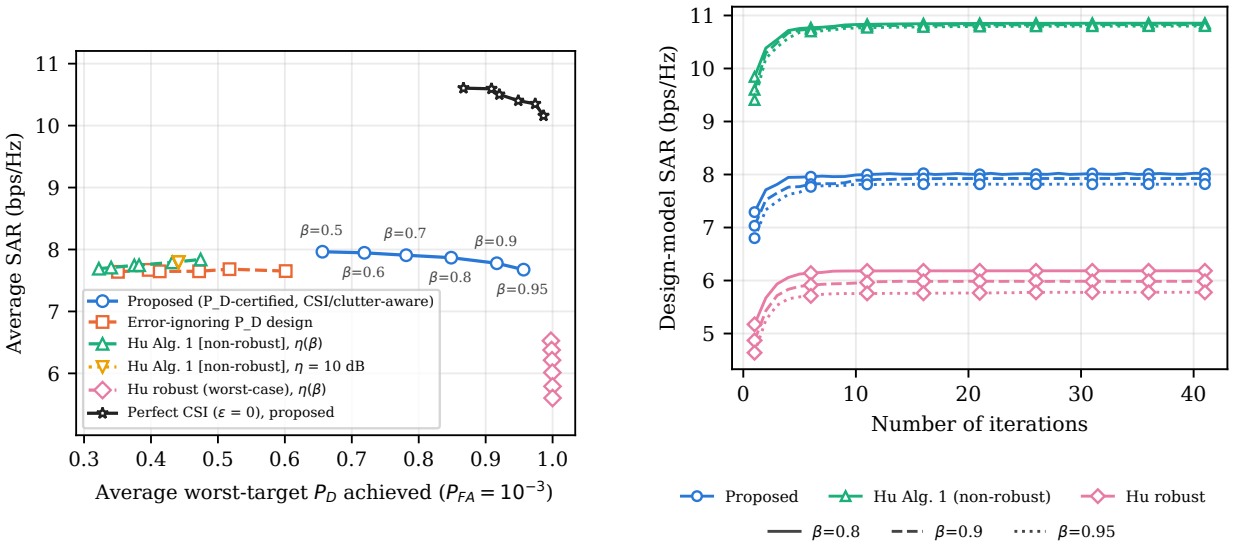


Figure 5: Left: rate versus achieved worst-target P_{D} along the β sweep (counterpart of [7, Fig. 6]). Right: convergence of the design-model rate (average of 30 channels); designs settle within 9.3 iterations on average.

Findings so far.

1. *CSI-error leakage is the dominant failure of SINR floors.* Protection of the non-robust designs falls from 70% at $\epsilon = 10^{-3}$ to 0% at $\epsilon = 0.1$ (Fig. 3).

2. *The clutter tail matters even with exact CSI.* At $\epsilon = 0$ the SINR floors protect 80–81% of channels against 100% for the proposed design.

3. *The advantage is largest under strong interference.* At $P_c = 25$ dBm the published floor protects no channel; with $N_r = 16$ the extra antennas suppress the leakage and both designs coincide (Table 6).

4. *Result Set 1 (single-pair relay).* Modelling the CSI error inside the detector doubles the certified ST power and raises the rate bound from 0.038 to 0.074 bit per channel use at identical held-out P_{D} and P_{FA}.

8 SOTA Positioning

Table 7: What existing approaches accomplish and how this project differs.

Approach	What it accomplishes	Difference in this project
SINR-floor cognitive relay [7]	Joint power and AF beamforming with sensing protection; robust worst-case extension	Protection is a calibrated P_{D} guarantee, valid under heavy-tailed clutter and CSI error; same architecture and setting, so directly comparable
Explicit P_{D} floor [18]	Shows that surrogate metrics can hide poor detection; adds a detection floor	Extends the explicit-detection idea to compound-Gaussian clutter, imperfect CSI and a cognitive relay
Compound-Gaussian ISAC detection [21, 19]	Exact detection analysis under realistic clutter	Uses that detection law as a *constraint* inside secondary-rate optimization
Clutter-aware / robust surrogates [9, 10, 11, 14]	SCNR or KLD optimization under clutter and uncertainty	Guarantees the radar specification itself rather than a surrogate
Worst-case robust design [7]	Always safe	Same safety with about 2 bps/Hz more protected rate (Table 5)

Contribution claims at M2. (i) A false-alarm-calibrated detection constraint for cognitive ISAC relaying under compound-Gaussian clutter and imperfect CSI; (ii) a monotonicity result that makes it an exact certificate inside a sum-rate design; (iii) a validated, paired comparison in the published setting of [7]. A web search of arXiv and open sources found no work combining these elements; a final IEEE Xplore and Scholar search is scheduled.

9 Limitations, Risks and Plan Towards M3

Table 8: Open items.

Item	Plan
Structured relays (ZF, MRC/MRT, LMMSE) forward the radar echo (88–93% of relay power at $P_c = 0$ dBm)	Add echo-cancelled versions for a fairer comparison
Convergence of the certificate-updating loop	Formal analysis; currently empirical
Radar beam fixed for all schemes	Joint optimization of $\mathbf{v}$
Number of secondary pairs fixed at $M = 4$	Sweep M to study scalability
Baselines are constraint sets, not the authors' code	Compare with a full GLDT-CS implementation if feasible
Novelty	Final IEEE Xplore / Scholar search before submission

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